

IMCA 221
Programming for Artists
Fall 2025

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**Class is on Zoom even in the classroom
for sharing, find the details on Moodle**

<https://moodle.concordia.ca/>

Download the slides!

Midi

MIDI stands for Musical Instrument Digital Interface, which is a technical standard that lets devices computers, musical instruments and other devices communicate.

Midi transmits data, NOT audio signals. Although your computer does have some default midi sounds.

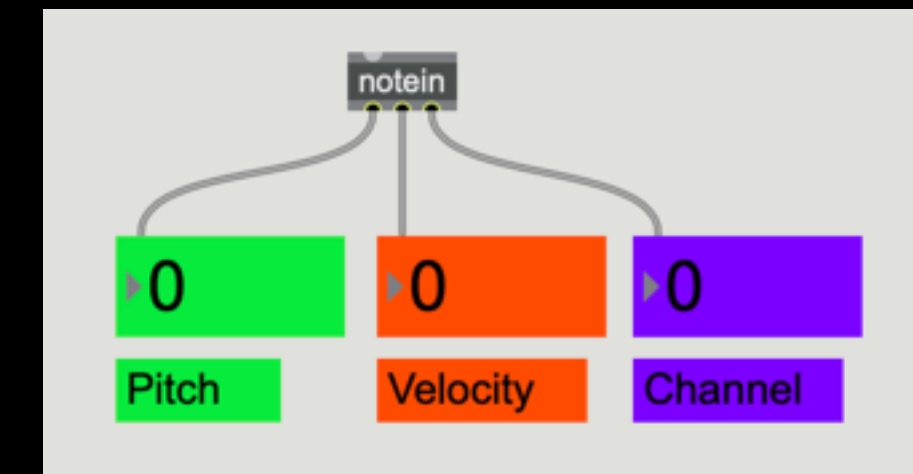
A midi message.

Midi messages have:

Pitch: A note number

Velocity: intensity

Channel: channel it is
being transmitted on



Midi Controller

Midi signals can come from external devices such as **midi controllers**. These provide midi signals you can use to interact with your patch such as a button, knob, keyboard, etc.

You can borrow midi controllers from the depot.

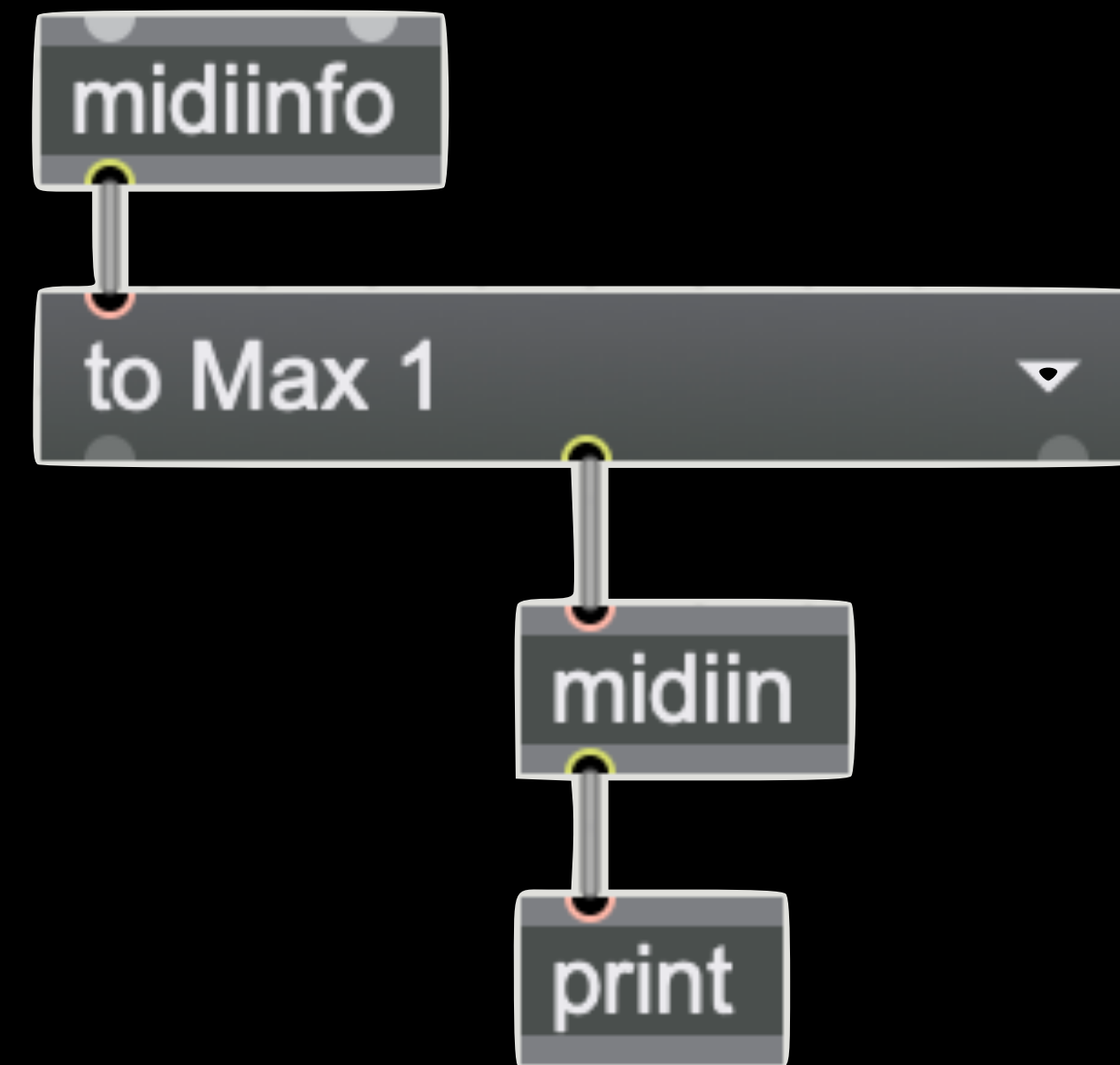


A midi message.
Plug in your midi
device. You will
need a midi to USB
cable (usually
comes with most
modern devices)



A midi message.

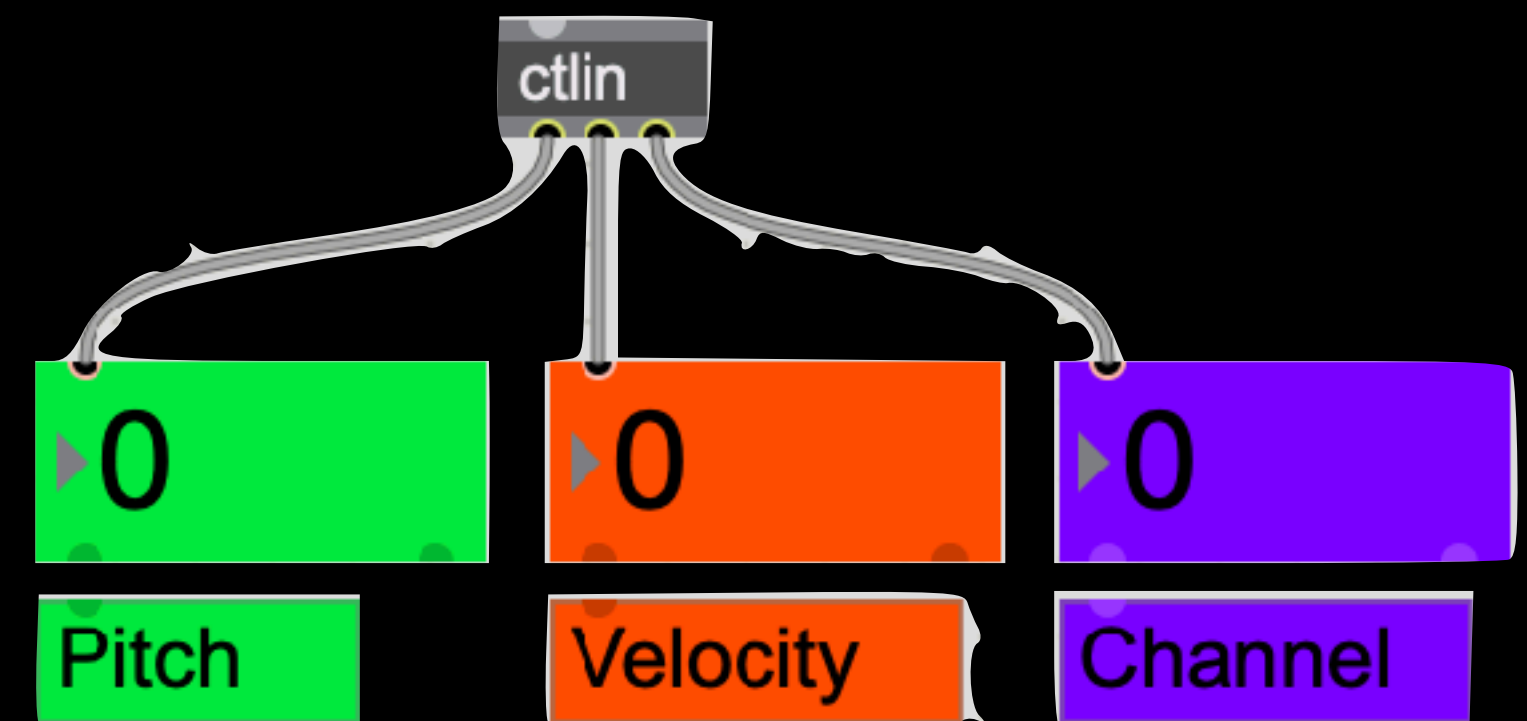
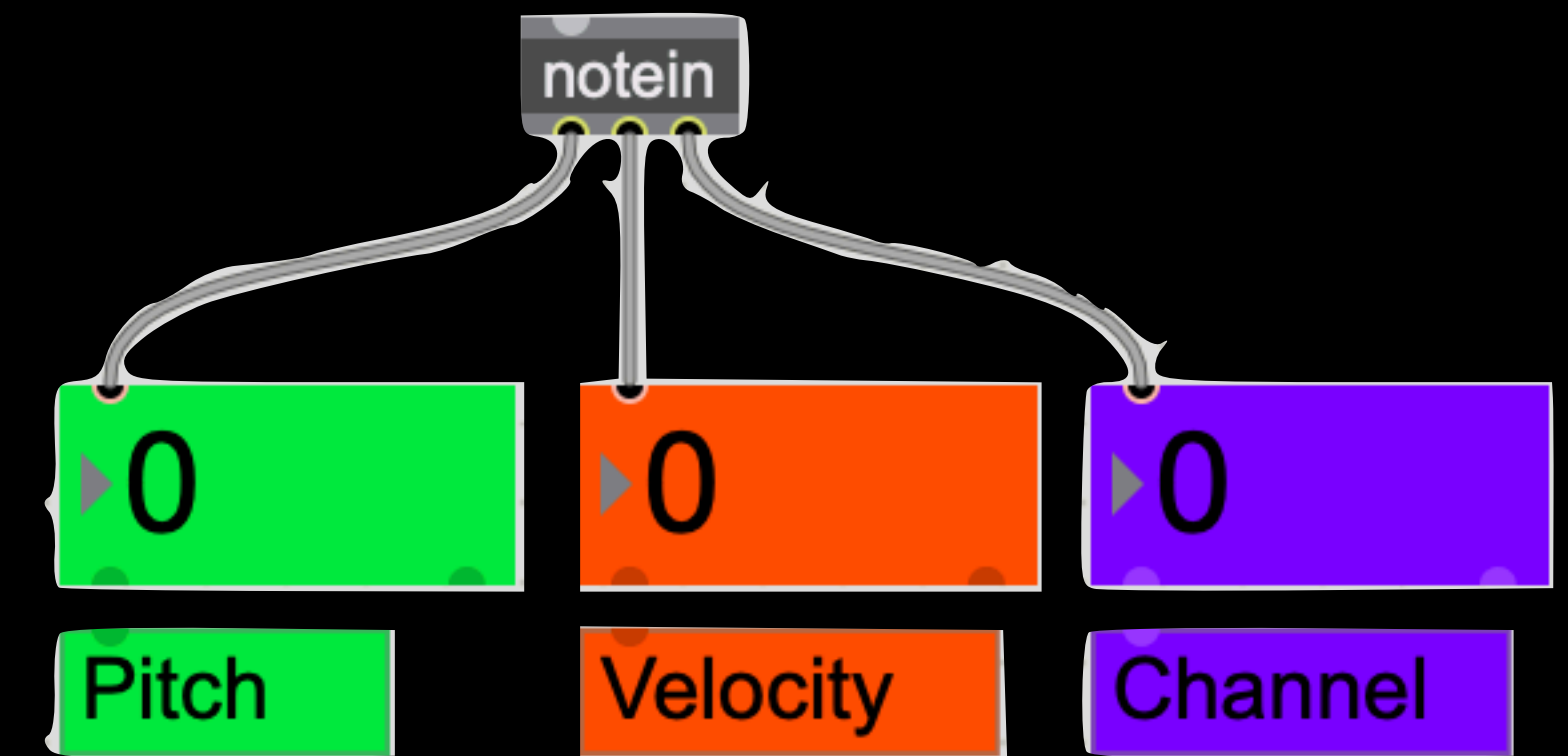
Use `midiinfo` object to view the raw midi info. This example prints it to the console. You may need to select the device from the dropdown.



A midi message.

Knobs / buttons will
often be **ctlin**
keys are often **notein**

You can use objects to
to view your midi
output.



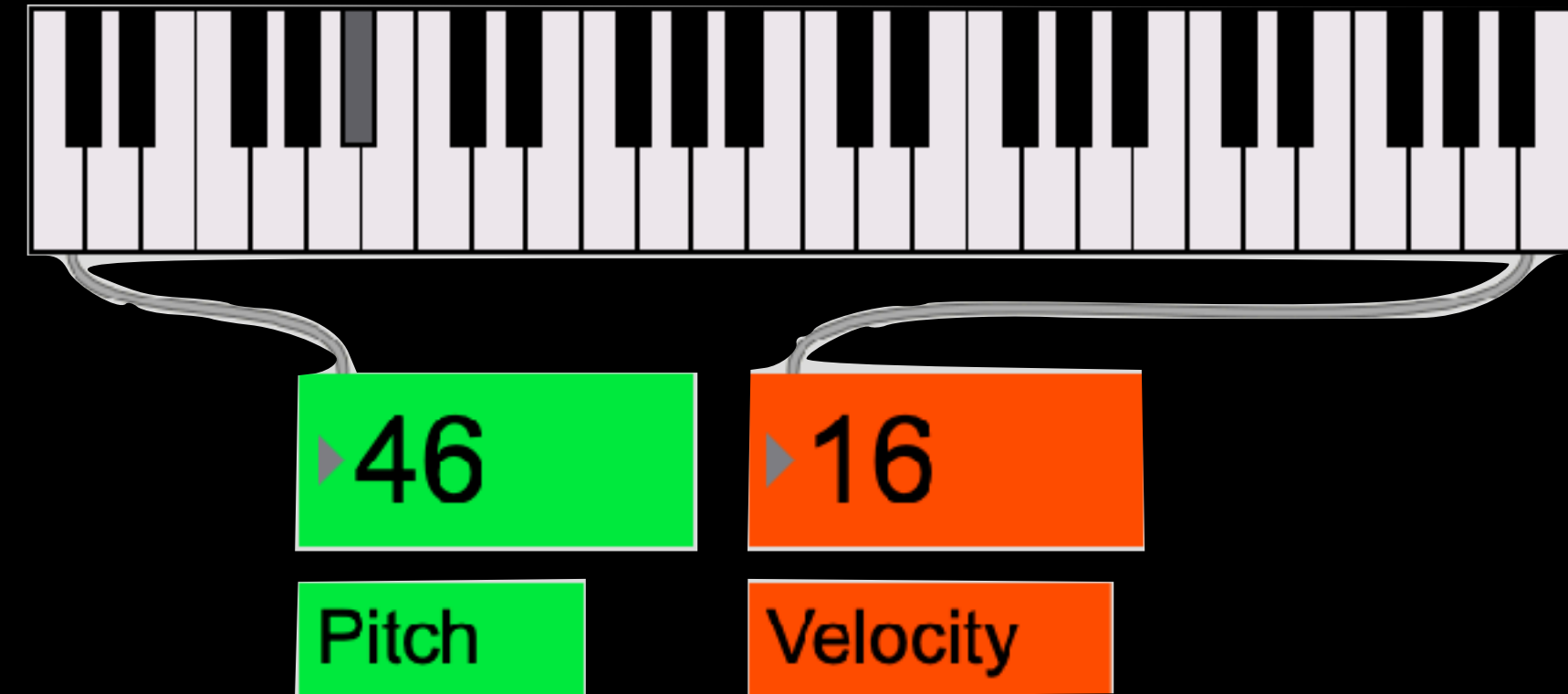
Ctlin

Use **ctlin** with a specific pitch/note number to monitor the value from a specific



Kslider

Kslider is a visual object for simulating a keyboard input.



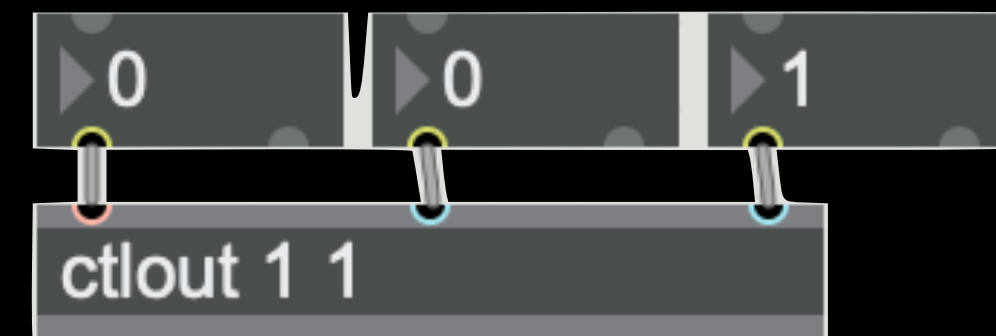
Midi Outputs

Midi can also output to control devices, such as lights. You can rent the **cyberpak** at the depot to control lights using midi.



Midi Outputs

Use ctlout to send data to specific midi channels to control things (such as cyberpak or midi power strip)



Midi mapping Ableton

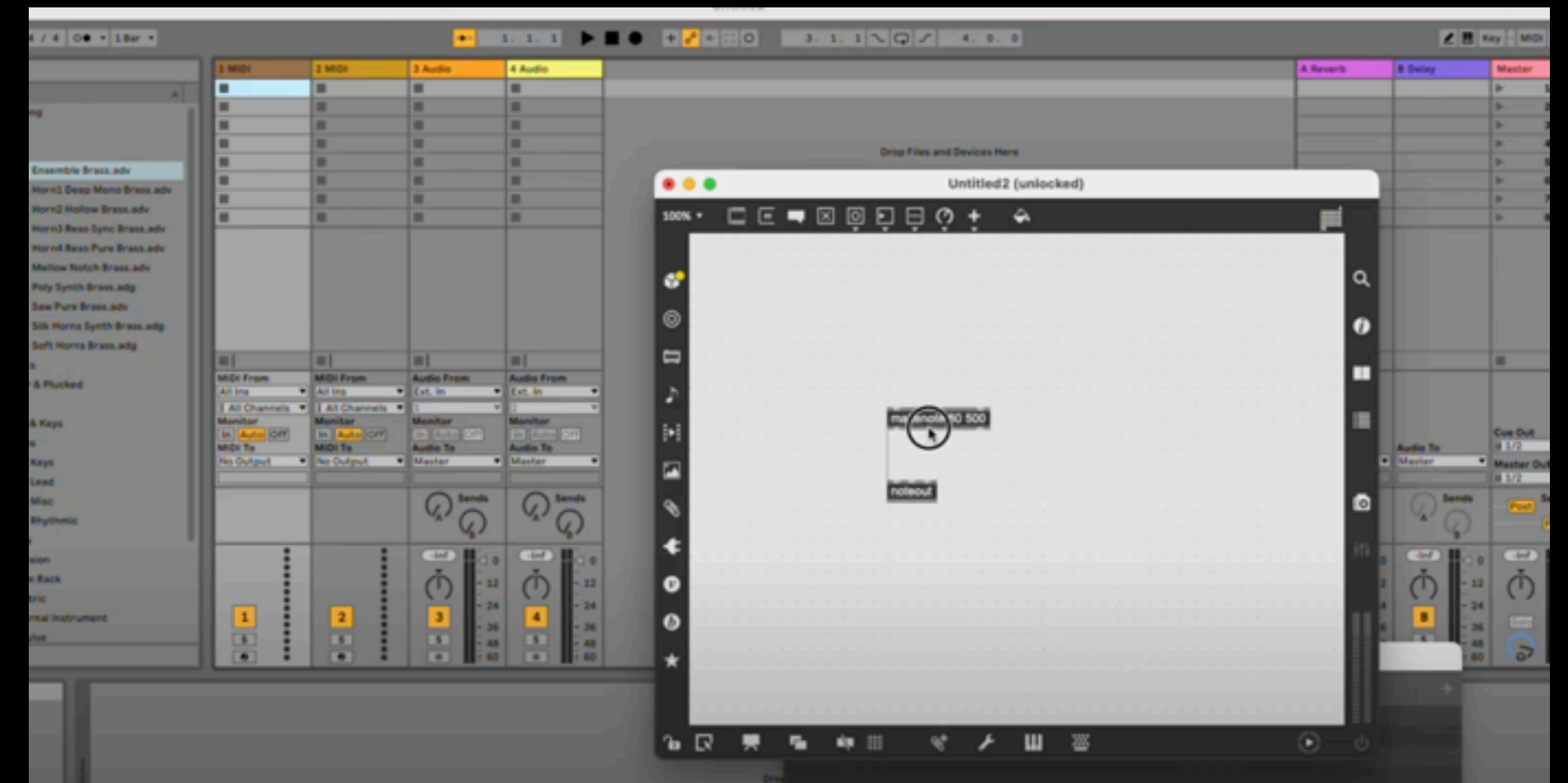
<https://www.youtube.com/watch?v=xmobq55Sg8w>

Ableton to Max:

<https://www.youtube.com/watch?v=kK76itbORSk>

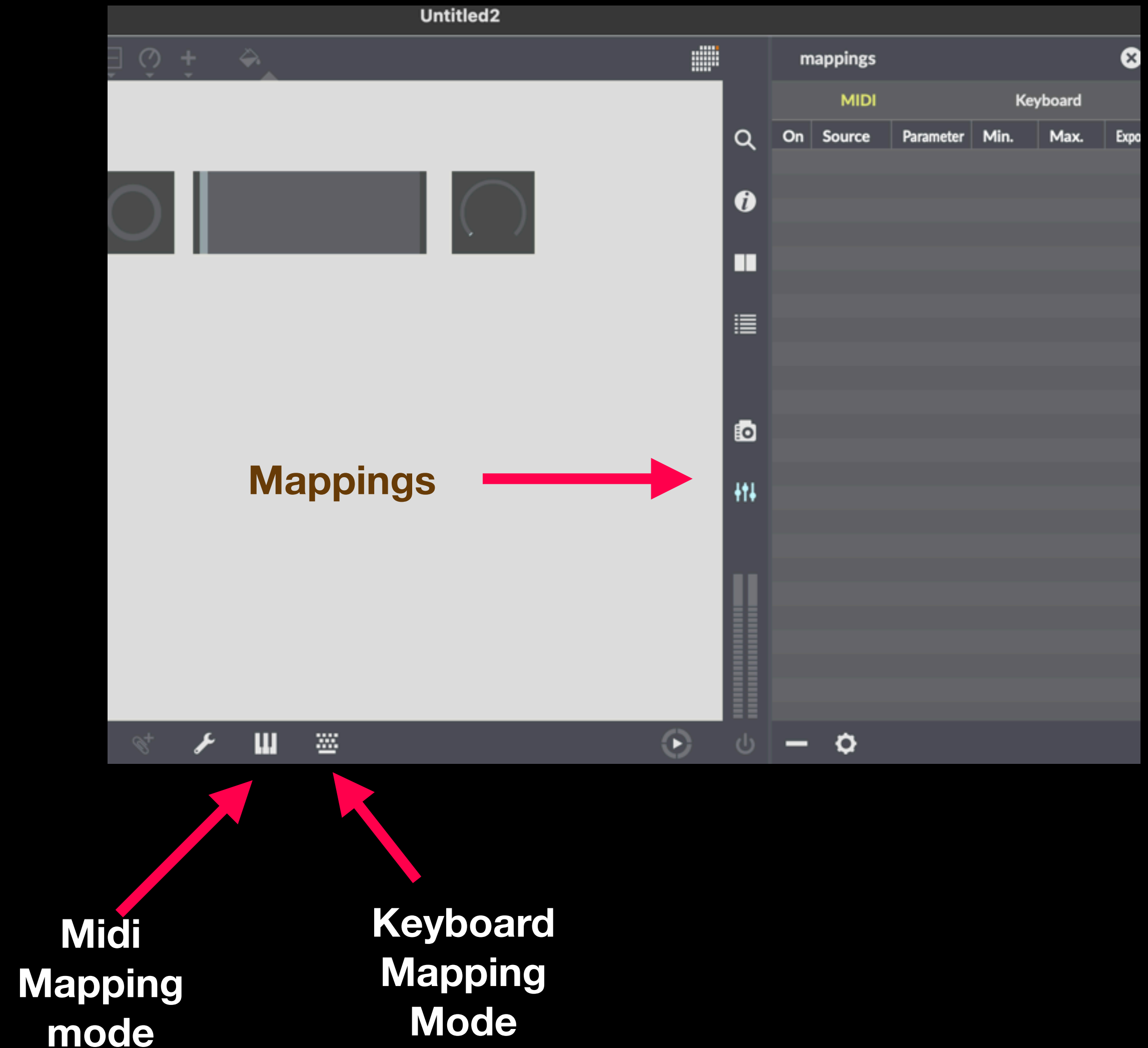
Max to Ableton:

<https://www.youtube.com/watch?v=RK0kVG3IZNQ>



Mapping

You can map keyboard or midi controls to objects using the mapping window.



Mapping

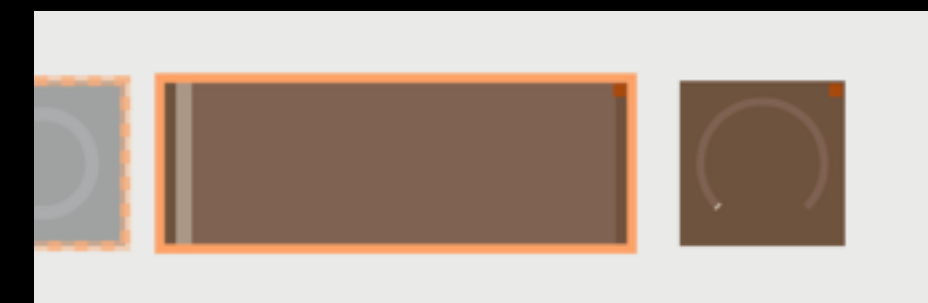
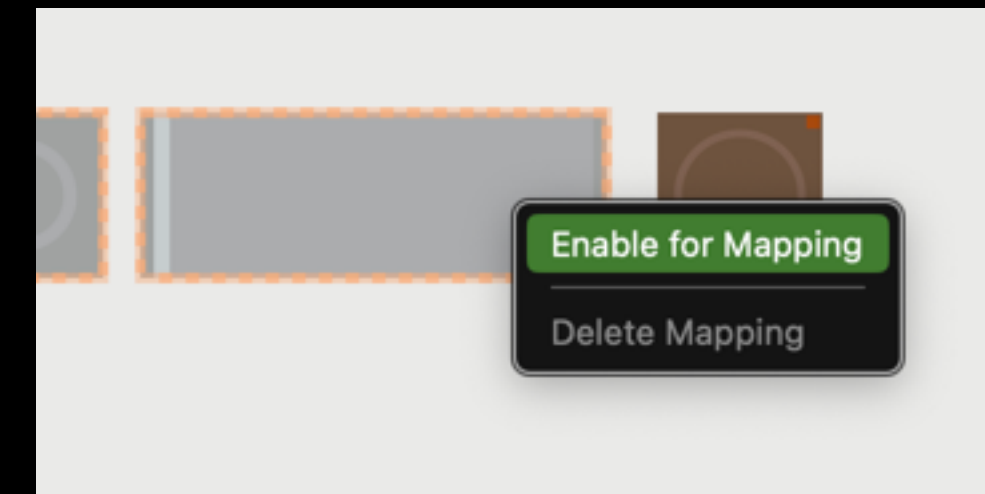
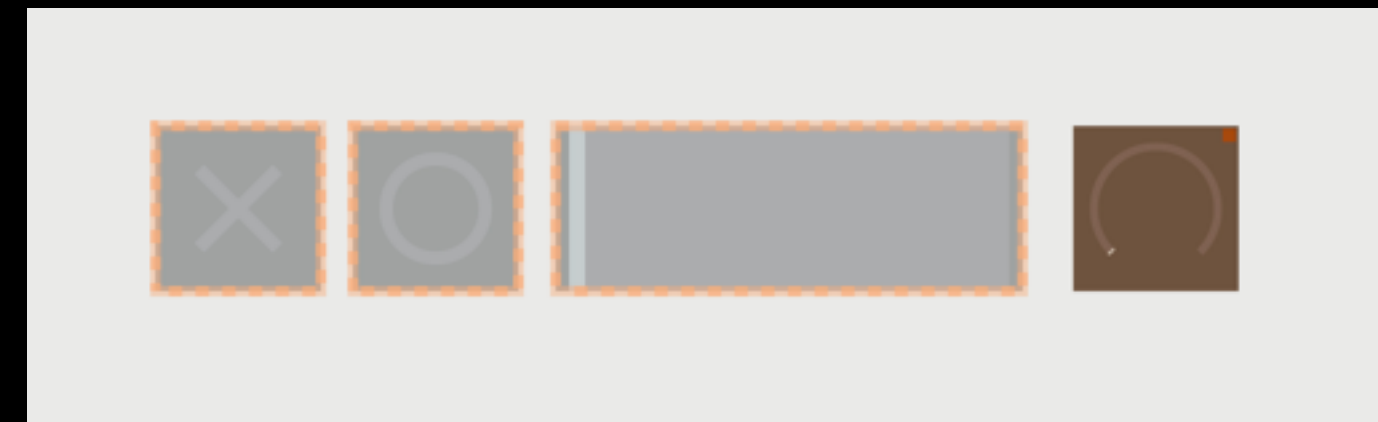
Enable keyboard or midi mapping mode

Objects able to be mapped will be outlined in purple/orange

Enable each object

Press the key you want to map to the object

View the mapping in the mappings window



Mapping

Use mappings to control UI elements wit Midi or keyboard

1. Open mappings panel (right tool bar, bottom icon)
2. Enable mapping (keyboard or midi) on bottom tool bar
3. Click item you want to map
4. Press key (or midi object) youw ant to use to control
5. View mapping in "mappings" panel and edit

See these mappings as examples.



Mapped to left toggle



Mapped to down, 127



Mapped to right momentary



Mapped to up, 50

Max.	Expon
1.000	1.000
127.000	1.000

Mapping

Use mappings to control UI elements with MIDI or keyboard

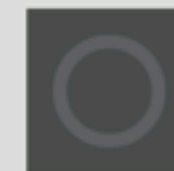
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Mapped to left toggle



Mapped to down, 127



Mapped to right
momentary



Mapped to up, 50

Mapping

mapping_midi_key
board maxpat

**Download today's patch
on Moodle**

Midterm: Transform - Groups of 2-3, 25%, October 21

In this project you are invited to look beyond the computer screen and transform a space or atmosphere. Using primarily audio and video (and other if you want to) MaxMSP tools, bring us into a transformed within (or nearby) our classroom. This could mean changing the mood, forcing us to change perspectives, or imagine something new. Your project should **not** appear as a max patch, but as a piece for people to approach and engage with through a linear progression or interactive element. Your work should in some way have an arch or experience that builds or changes throughout the experience.

Be intentional. What do you want us to experience, and how are you bringing us there?

This work should feel complete - take note of where wires and cables are. Use projectors, lights, staging, curation etc to create a space. The work should be approaching gallery-ready state. Feel free to re-arrange any nearby space or reserve a critique room.

Homework:

Work on Midterm project

Next class: Critique!